

Yashas M Shetty

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EDUCATION

New Horizon College of Engineering

B.E. in Artificial Intelligence and Machine Learning

Bangalore, KA

May 2026

- **GPA:** 8.47/10.0
- **Related Coursework:** Machine Learning, Deep Learning, NLP, Artificial Intelligence, Gen AI, Agentic AI

SKILLS

AI / Agents: Google ADK, LLM Agents, Multi-Agent Systems, RAG, Prompt Engineering, Vertex AI, Agent Memory

Frameworks & Libraries: FastAPI, React, LangChain, PyTorch, TensorFlow, Scikit-learn, Streamlit

Languages: Python, Java, JavaScript, SQL

Databases & Infra: PostgreSQL, FAISS, ChromaDB, Docker, Git

EXPERIENCE

MakeMyTrip (redBus) - [ADK, LLM Agents, Java]

Bangalore, KA

Software Engineering Intern.

Sep 2025 – May 2026

- Built and integrated **LLM-powered chatbot agents** using Google **ADK**, improving automation for internal workflows.
- Designed artifact persistence infrastructure for the **Ray chatbot**
- (live in the redBus app), enabling storage of BiDi voice interactions and STT artifacts to power downstream evaluation pipelines, debugging, and model behavior analysis
- Contributed to modular **agent workflows** and developed **custom tools/APIs** for enhanced agent capabilities.
- Contributed to **adk-java**, an open-source fork of Google ADK, by building **PostgresArtifactService** and **PostgresArtifactStore** — replacing in-memory/GCS-only persistence with production-grade PostgreSQL-backed storage for multimodal artifacts (audio, images, documents) Github: <https://github.com/redbus-labs/adk-java/>
- Developed a **File Analyzer Agent** and artifact inspection tools to dynamically parse and evaluate files, videos, images and docs.
- Engineered guardrails for HTTP/WebSocket pipelines to mitigate prompt injection, sanitize inputs, and enforce token limits — hardening AI interaction safety for production-scale user traffic.
- Built an amenity detection system using **Gemini 2.5 Flash** and **ADK** to automatically classify bus amenities (AC, blanket, charging point, etc.) from images — eliminating the need for manual image review and enabling operators to upload images and instantly get structured amenity data for listings.

PROJECTS

Proximity detection and safety zone monitoring using computer vision - [YOLOV5]

Bangalore, KA

Team Project

May 2024 – Aug 2024

- Developed a computer vision-based system to enhance site safety by monitoring activities and ensuring workers do not accidentally enter dangerous/hazardous zones in factories and construction sites.
- Utilized YOLOv5 to implement proximity detection, preventing unauthorized entry into restricted or hazardous areas by tracking workers and using virtual hazardous zones using ROI and bounding boxes.
- This project uses real-time image recognition to detect safety violations and generates instant alerts for corrective action, reducing the risk of accidents, and fostering a safer work environment.

RAG Chatbot - [Ollama, FAISS, Huggingface, FastAPI]

Bangalore, KA

Team Project

May 2024 – Jun 2024

- Built a Retrieval-Augmented Generation chatbot that answers queries from uploaded PDFs using semantic search with FAISS and HuggingFace embeddings.
- Integrated Ollama LLM (Llama 3.2:3B) via LangChain with conversational memory and custom prompt templates for summarization and reasoning.
- Developed a FastAPI backend for vector indexing and a Streamlit frontend for interactive chat with real-time Q&A.

ACHIEVEMENTS AND CONTRIBUTIONS

- **Research Paper (ICICC 2025):** System and Method for Virtual Boundary Detection and Warning of Safety Zone Violations in Construction and Industrial Environments - **Paper ID: 1346**